

Limitations of the Trajectory & Bowling-Machine Algorithm

What the model gets right, and where it's known to fall short

A companion note to “The Physics of a Cricket Ball in Flight” — read alongside that report's Section 8 (calibration) and Section 10 (caveats).

12 September 2026

This model turns cricket-ball aerodynamics into a runnable trajectory simulator and a bowling-machine control mapping. It's built to be honest about where it's solid and where it's approximate — this note collects every known limitation in one place, roughly in order of how much they matter in practice.

1. The aerodynamic coefficients are fitted, not measured

$C_d(\text{Re})$, $C_l(\text{Sp})$, $C_s(\theta)$ and $C_{ds}(\theta)$ are all curves shaped to reproduce the right qualitative behaviour — a drag crisis near bowling Reynolds numbers, a swing peak near 22° , a saturating Magnus lift — rather than numbers measured from wind-tunnel tests on the actual ball in use. Until the calibration steps are done, the model is illustrative rather than predictive. This is the single biggest source of error, and the one most worth fixing first.

2. No turbulence physics is actually simulated

The boundary-layer transition that causes both the drag crisis and swing is only described in prose. Nothing in the algorithm solves Navier-Stokes or runs any turbulence model — no RANS with a $k\text{-}\epsilon/k\text{-}\omega$ closure, no LES, no DNS. The curves stand in for that physics; they don't derive it from first principles.

3. Swing is frozen at release

The model assumes the seam holds one fixed angle for the entire flight. Real deliveries wobble slightly in the air — no bowler holds the seam mechanically perfectly still — which is exactly why real swing sometimes looks inconsistent ball to ball even at the same nominal speed and seam angle. The model can only give one deterministic answer per set of inputs; it can't reproduce that delivery-to-delivery variability.

4. Spin is treated as constant

Air resistance applies a small decelerating torque to a spinning ball, so real spin rate decays slightly over the flight. This model keeps spin rate fixed from release to pitching — a reasonable approximation over roughly half a second of flight, but not exactly zero error.

5. Reverse swing is a toggle, not a mechanism

Past the pace threshold (140 km/h by default), the swing coefficient simply gets multiplied by a fixed gain. The real roughness-asymmetry mechanism behind reverse swing is still active research territory in sports aerodynamics, so this was never going to be more than a simplified stand-in.

6. The swing/seam-drag split is a geometric idealisation

Resolving one hypothetical seam-generated force into a $\cos(\theta)$ swing component and a $\sin(\theta)$ drag component (Sections 3.4-3.5 of the main report) is more physically motivated than fitting one lateral curve directly — but it's still one idealised force resolved two ways, with a tuned peak-shape exponent (n) and a separately tuned $C_{ds,max}$, not two independently validated experimental curves.

7. The machine release model assumes idealised wheel contact

The twin-wheel model assumes two wheels with single, constant efficiency factors for speed transfer (η_v) and spin transfer (η_ω), with no dependence on wheel wear, ball condition, or speed regime, and no direct control over seam angle beyond whatever a separate feeder mechanism sets. A three-wheel machine, or one with meaningfully different contact geometry, needs its own version of that section.

8. Nothing after pitching is modelled

The simulation stops the instant the ball reaches ground level. Bounce height, seam grip, friction with the pitch surface, and the post-bounce trajectory on to the bat are entirely outside this algorithm's scope — it only covers release to first ground contact.

9. The environment is simplified

Air density is a single constant per run (with a rough exponential correction for altitude), wind is a fixed vector rather than anything gusty or turbulent, and temperature/humidity effects on air properties are only present insofar as they're baked into whatever density and viscosity values are set by hand.

Practical priority if you're calibrating a real machine: fix η_v and η_ω first (cheapest to measure, biggest accuracy win on pace and line), then fit the drag curve against a radar gun if predicting pace matters, and only chase the swing/spin curve calibration once those are solid — it's the most expensive to measure accurately and the least forgiving of noisy data.